

Machine Learning: Product Review Sentiment Analysis

A Comparative Study of Classification

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SERVICES



IDEA



TEAM



Statement



GOAL

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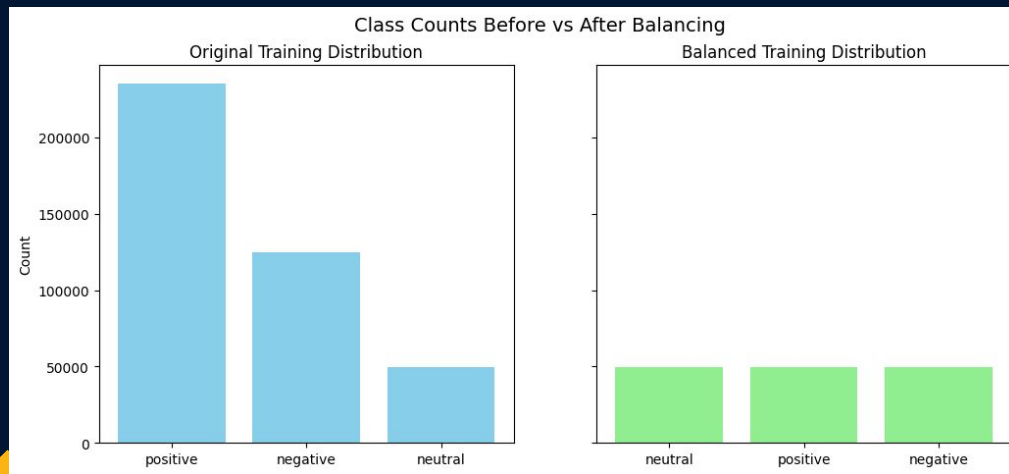
02

Fisher

Mustafa Abou Khatwa

Motivation

- Severe class imbalance (Neutral underrepresented, Positive dominant).
- Need for interpretable linear baseline alongside Logistic Regression.
- Goal: Evaluate Fisher's discriminant with hybrid features and balancing strategies.



Methodology Overview



- 1 Data Preparation
 - a. Balanced training set: 49,373 samples per class.
 - b. Lexicon-only (VADER scores) vs Hybrid (TF-IDF + lexicon).
- 2 Model Design
 - a. Custom FisherEstimator with SVD dimensionality reduction.
 - b. Priors adjusted via prior_alpha (inverse-frequency weighting).
 - c. Optional Neutral upsampling capped at 50,000.
- 3 Tuning
 - a. GridSearchCV for solver, shrinkage, priors, and dimensionality.
 - b. Calibration with isotonic regression + threshold sweeping.

Results

Lexicon vs Hybrid

01

Lexicon-only

Accuracy: 0.787

Macro-F1: 0.439

Neutral F1: 0.085

02

Hybrid

Accuracy: 0.734

Macro-F1: 0.458

Neutral F1: 0.169

03

Key Insight:

Hybrid features consistently outperform lexicon-only, especially for Neutral sentiment.

Results

Lexicon vs Hybrid

	Accuracy	Macro Precision	Macro Recall	Macro F1	Micro F1	ROC-AUC	F1-negative	F1-neutral	F1-positive	CV Mean ± Std
Lexicon-only	0.787 	0.462	0.427	0.439	0.787 	0.719 	0.346	0.085	0.886 	nan
Hybrid	0.734	0.465 	0.461 	0.458 	0.734	0.718	0.352 	0.169 	0.852	nan

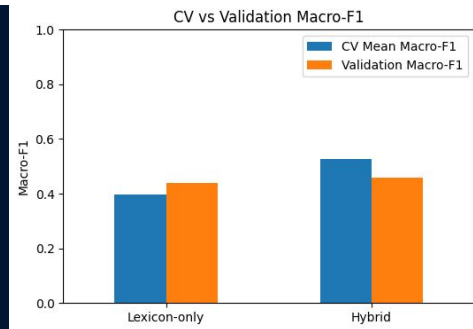
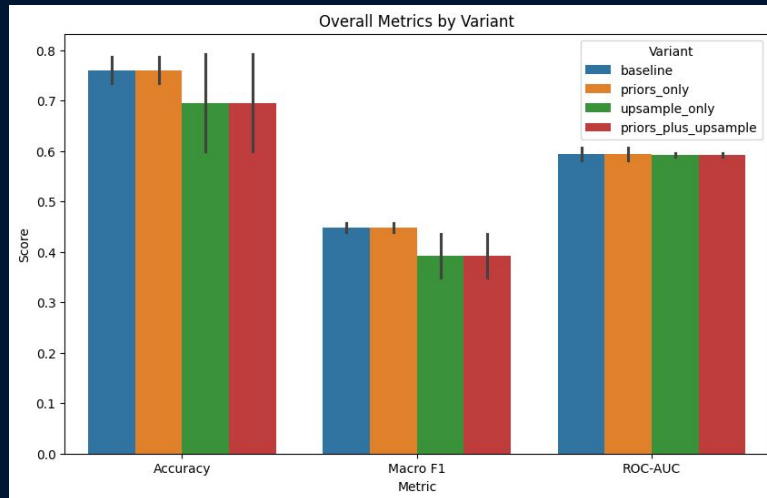
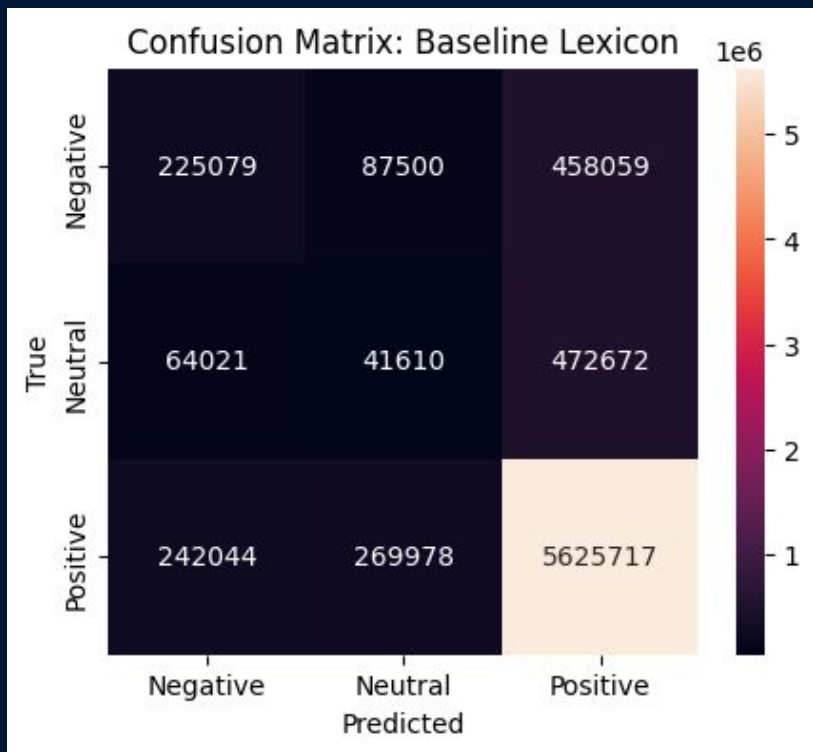
Results

Hyperparameter Tuning

1. Best config: `svd_n_components=100`, `lda_solver=lsqr`, `lda_shrinkage=None`, `prior_alpha=0.5`.
2. Validation Macro-F1: 0.453
3. Neutral F1: 0.125
4. CV Macro-F1: 0.490 (stable generalization).
5. Takeaway: Dimensionality reduction + tempered priors yield balanced performance.



Validation & Diagnostics



Challenges and Responses / Future Directions

Problem: Neutral class performance remains weak.

Response/Future Direction:

Acknowledge linguistic overlap; calibration and upsampling partially mitigate. Explore contextual models (e.g., BERT, RoBERTa) to better capture semantics

Problem: Linear models underperform compared to Neural Networks.

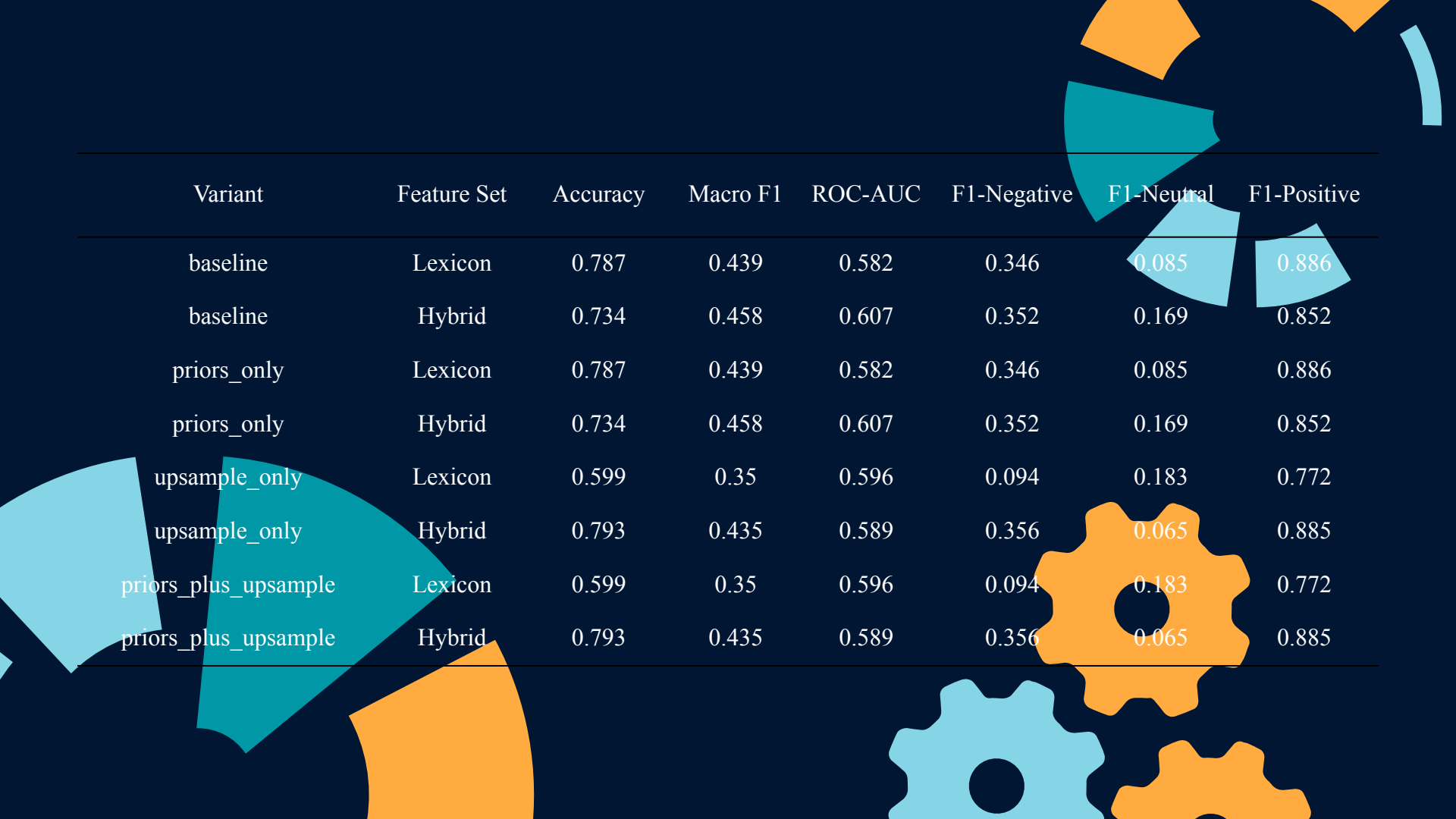
Response/Future

Direction: Emphasize interpretability, reproducibility, and auditability — critical in regulated contexts. Future work: adopt deep learning for complex boundaries while retaining linear baselines for audit -ready pipelines.

Problem: Priors and upsampling effects are limited.

Response/Future

Direction: Show that hybrid features + dimensionality reduction are the real drivers of improvement. Future work: hierarchical classification to separate Neutral from Positive/Negative more effectively.



Variant	Feature Set	Accuracy	Macro F1	ROC-AUC	F1-Negative	F1-Neutral	F1-Positive
baseline	Lexicon	0.787	0.439	0.582	0.346	0.085	0.886
baseline	Hybrid	0.734	0.458	0.607	0.352	0.169	0.852
priors_only	Lexicon	0.787	0.439	0.582	0.346	0.085	0.886
priors_only	Hybrid	0.734	0.458	0.607	0.352	0.169	0.852
upsample_only	Lexicon	0.599	0.35	0.596	0.094	0.183	0.772
upsample_only	Hybrid	0.793	0.435	0.589	0.356	0.065	0.885
priors_plus_upsample	Lexicon	0.599	0.35	0.596	0.094	0.183	0.772
priors_plus_upsample	Hybrid	0.793	0.435	0.589	0.356	0.065	0.885

Conclusion

Fisher's Linear Discriminant is a defensible, interpretable Baseline.

Hybrid features consistently outperform lexicon-only.
Calibration and upsampling highlight trade-offs in imbalance Handling.

Provides audit-ready reproducibility, even if not the top-performing model.

02

Logistic Regression

Mohamed Mourad

Model Design



- Algorithm: Logistic Regression for multiclass sentiment classification
- Regularization:
L2 (ridge): stability & generalization
L1 (lasso): sparsity & feature selection
- Class handling:
class_weight = balanced
- Solver:
lbfgs (L2)
saga (L1)

Features and Training

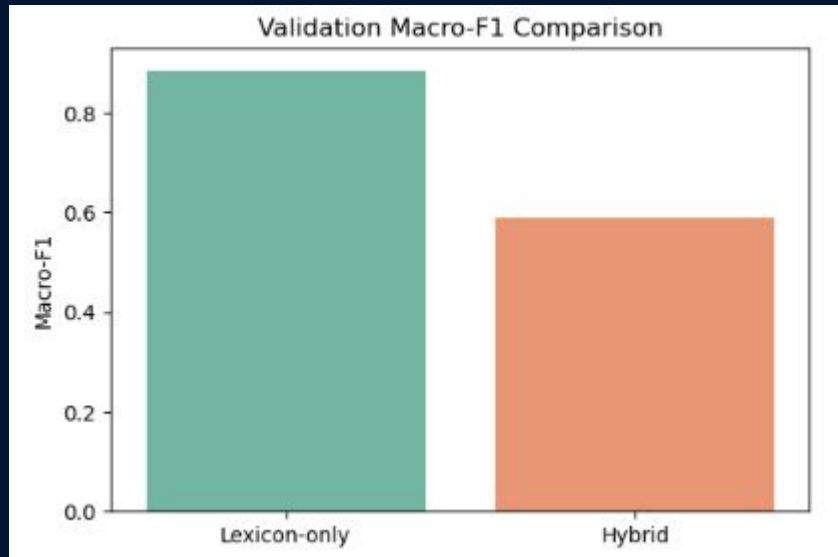
- Lexicon-only: VADER sentiment scores
- Hybrid: TF-IDF + sentiment lexicons
- Training strategy:
 - Balanced labels
 - GridSearchCV for C and penalty
 - Macro-F1 as primary metric
- Evaluation Focus:
 - Fairness across all sentiment classes.



Results and Takaways

- lexicon-only features outperform Hybrid
- Strong performance on Positive & Negative
- Neutral class remains challenging
- Key strengths:
 - Interpretability
 - Reproducibility
 - Audit-ready baseline
- Conclusion:

Logistic Regression is a reliable, transparent baseline, not the strongest model, but essential for understanding gains from more advanced methods.



03

Perceptron

Ahmed Gomaa

Perceptron Methodology

1

Feature Augmentation: Manual calculation of interaction features.

2

Strategic Undersampling: Balancing the training data distribution.

3

Model Architecture: Calibrated Perceptron within a robust pipeline.

4

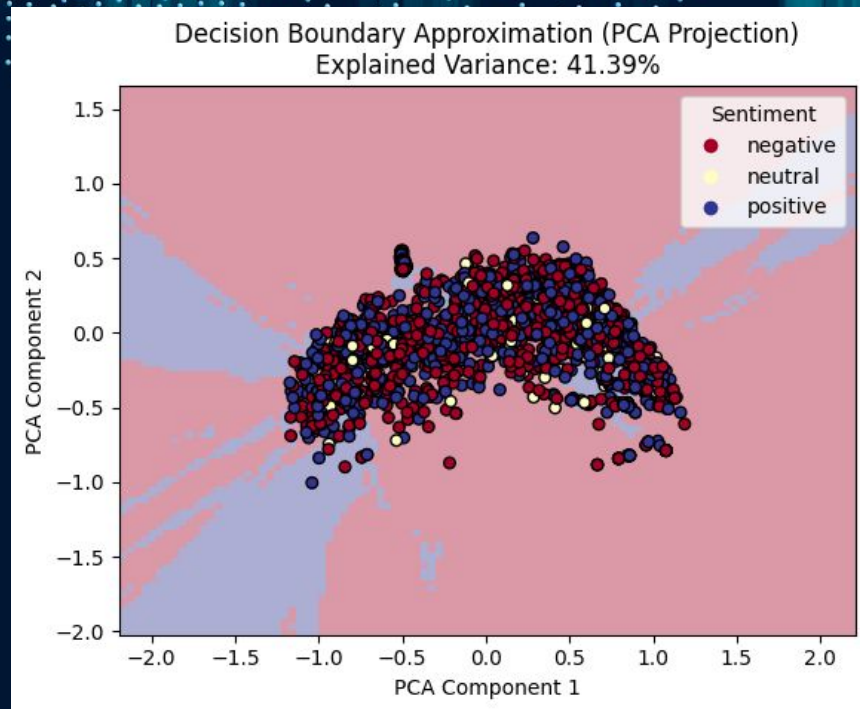
Ensemble Learning: Bagging stabilization.

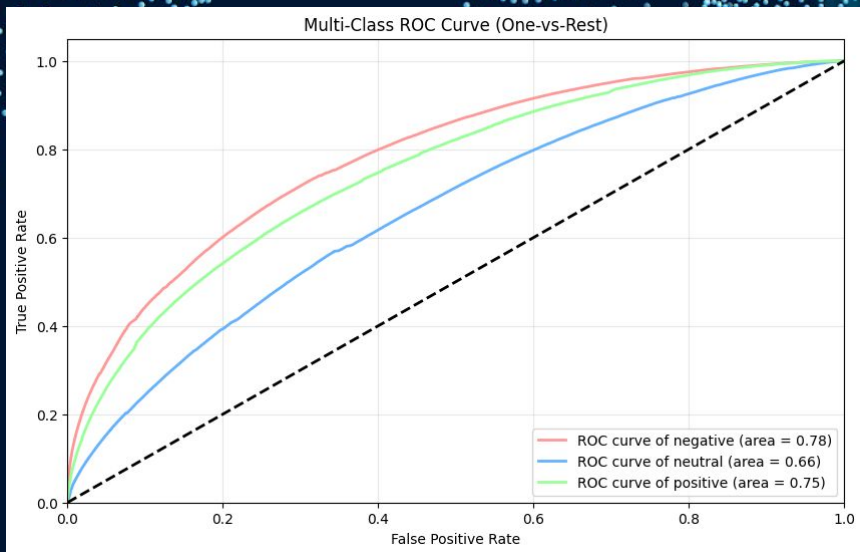
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Threshold Optimization: Post-hoc tuning for minority class sensitivity

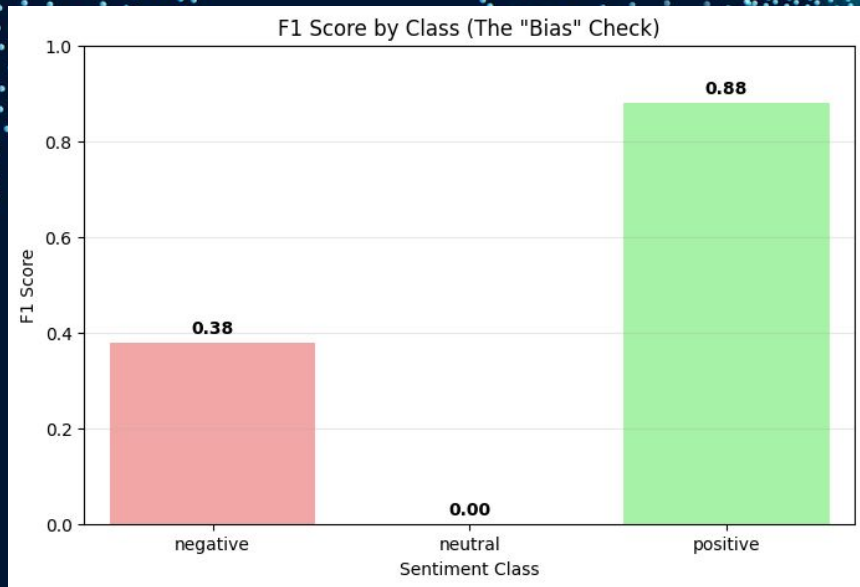


PCA OF THE DATA





Beyond Accuracy: Measuring Predictive Potential
with ROC



The Bais Check

Feature Space Augmentation



Goal: Help linear models detect non-linear relationships (mixed/neutral sentiment).

1

Feature Augmentation: Manual calculation of interaction features.

2

Strategic Undersampling: Balancing the training data distribution.

3

Model Architecture: Calibrated Perceptron within a robust pipeline.

4

Threshold Optimization: Post-hoc tuning for minority class sensitivity.



Strategic Undersampling & Architecture

Class Balancing

- **Minority:** Retained 100% of "Neutral" samples.
- **Majority:** Randomly downsampled "Positive" and "Negative" to 2 times the size of Neutral.
- *Result:* Loss function weights minority classes sufficiently.

Pipeline Components

1

Selection: `SelectKBest` (ANOVA F-value) to remove noise.

2

Scaling: `StandardScaler` for stable linear convergence.
Classification: `Perceptron` (SGD-based) wrapped in

3

`CalibratedClassifierCV` for probabilistic outputs.



Ensemble & Optimization

Ensemble Stabilization (Bagging)

- **Technique:** Bootstrap Aggregating with 10 independent estimators.
- **Training:** Each estimator sees a random 70% subset of balanced data.
- **Output:** Averaged probabilities to reduce variance.

Decision Threshold Optimization

- **Problem:** Standard "Argmax" favors the majority class.
- **Solution:** Grid search for optimal probability thresholds on the validation set.
- **Goal:** Maximize Macro F1-Score, not Accuracy.



Results - Baseline vs. Hybrid

1. Baseline (Lexicon-Only)

- **The Accuracy Paradox:** 81.9% Accuracy, but **0.00 Recall** for Neutral/Negative.
- **Function:** Acted as a trivial "Positive Detector."

2. Hybrid Model (Feature Engineering)

- **Interaction Features:** Successfully "unlocked" Negative sentiment.
- **Improvement:** Negative Recall rose to **0.18**.
- **Limitation:** Neutral class remained undetected (Recall 0.00).



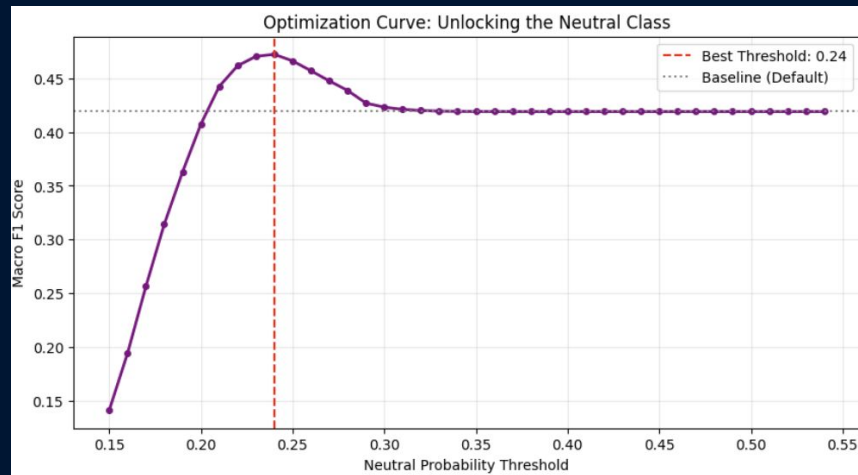
Results - Final Optimized Ensemble

The Breakthrough

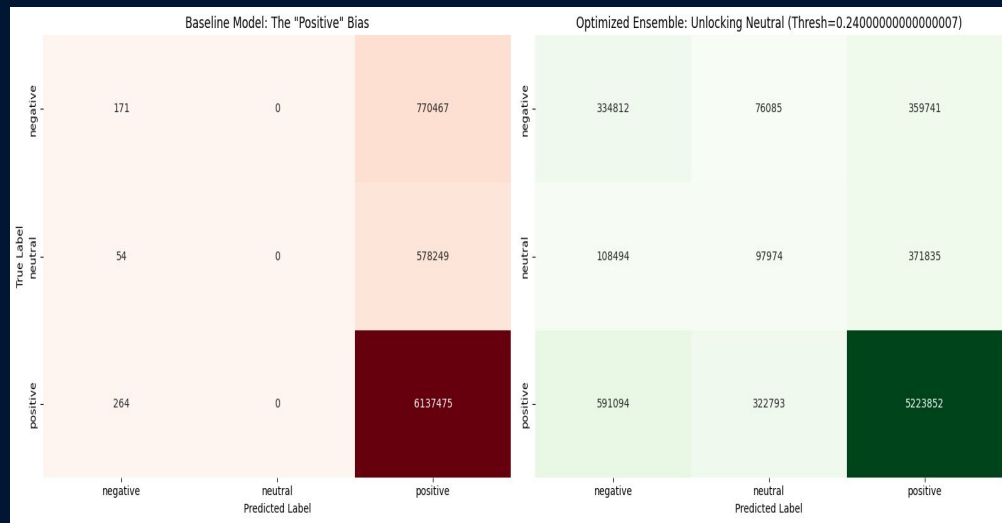
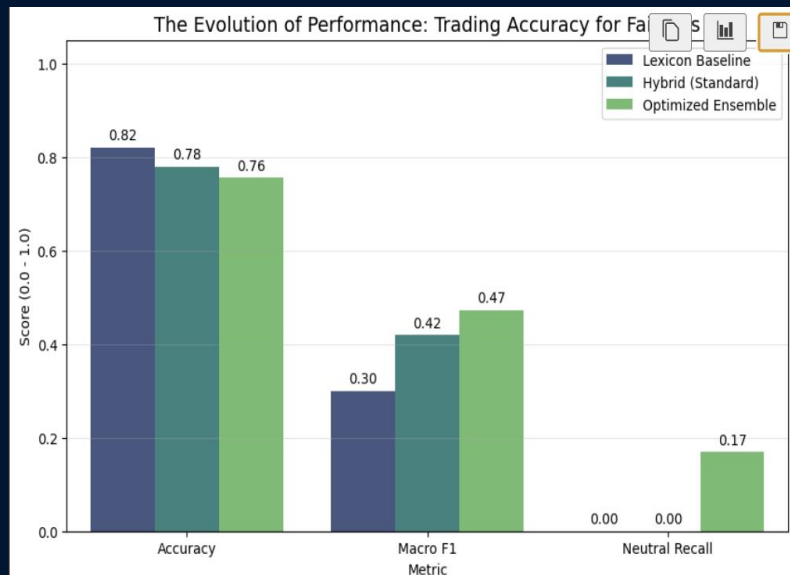
- **Macro F1-Score:** Peaked at **0.47** (56% improvement over baseline).
- **Unlocking Neutral:** Threshold tuning raised Neutral Recall to **0.17** (previously 0).
- **Negative Detection:** Massive improvement to **0.43 Recall**.

The Trade-off

- **Accuracy:** Dropped from 82% to **76%**.
- **Positive Recall:** Dropped from 98% to **86%**.
- *Conclusion:* We traded superficial accuracy for algorithmic fairness.



Performance Comparison



04

Support Vector Machine (SVM)

Amira Ibrahim Elshehaby

Enhancing SVM Sentiment Classification through Hybrid Resampling and Stochastic Optimization

Key Challenge 1:

Large-scale, imbalanced sentiment data

Aim

- ❑ Improve minority class detection (Neutral & Negative)
- ❑ Maintain scalability for very large datasets
- ❑ Compare linear, hybrid, ensemble, and deep models



Severe Class Imbalance

Severe class imbalance

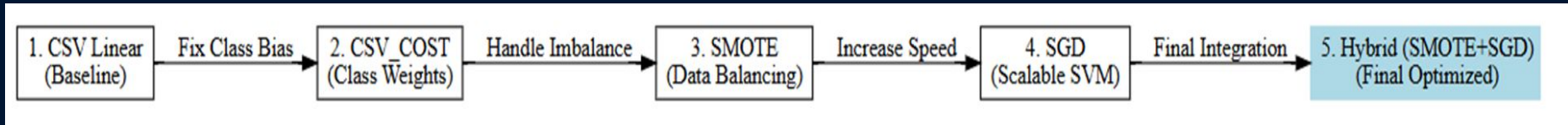
- ❑ Positive: **81.9%**
- ❑ Neutral & Negative: minority classes

Baseline SVM suffered from **Neutral Collapse**

- ❑ Neutral Recall = **0.00**



Pipeline of Improvement

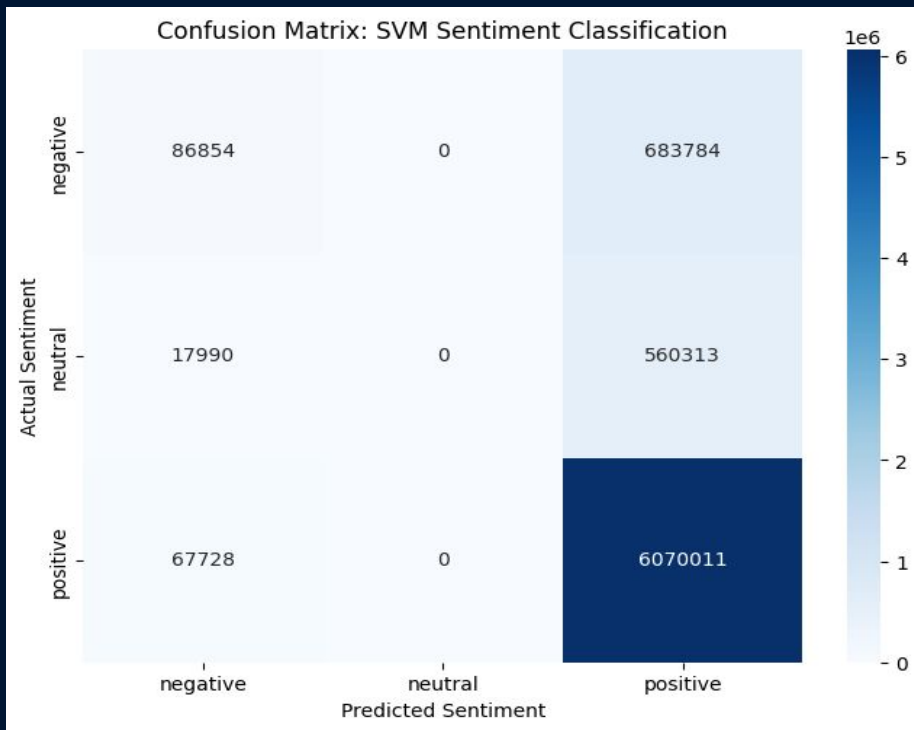


Methodology Overview

Iterative Improvement Pipeline :

- ❑ Baseline Linear SVM
- ❑ Cost-Sensitive Learning
- ❑ Data Balancing (SMOTE)
- ❑ Scalable Optimization (SGD)
- ❑ Hybrid Model (SMOTE + SGD)
- ❑ Model Performance Comparison: Baseline SVM vs. Optimized Hybrid
- ❑ Deep Learning

Baseline Model (CSV Linear)



➤ Method:

Linear Support Vector Classifier

➤ Result:

- High overall accuracy
- Neutral Recall = 0.00

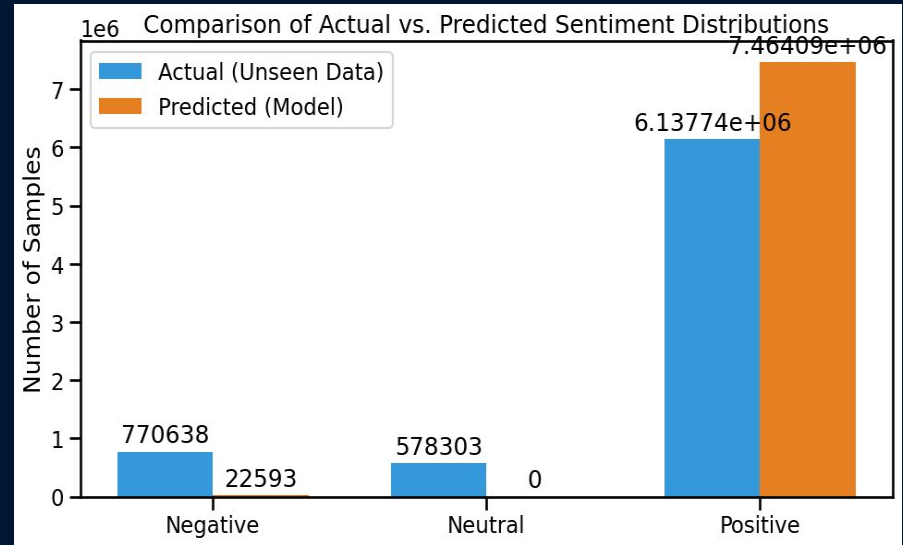
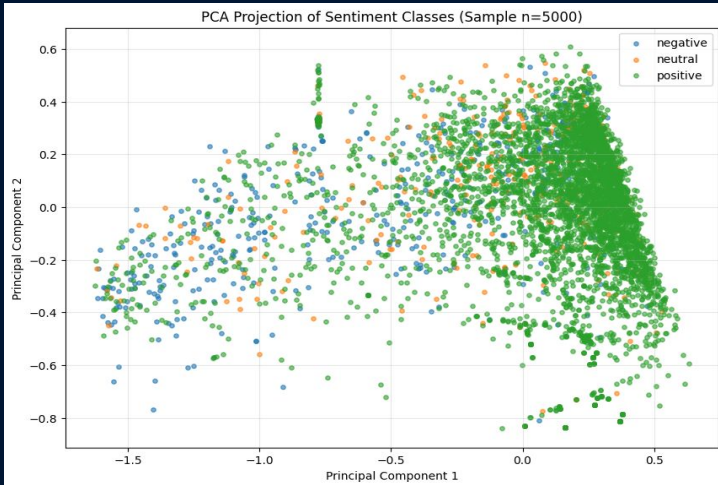
➤ Issue:

Strong majority class bias

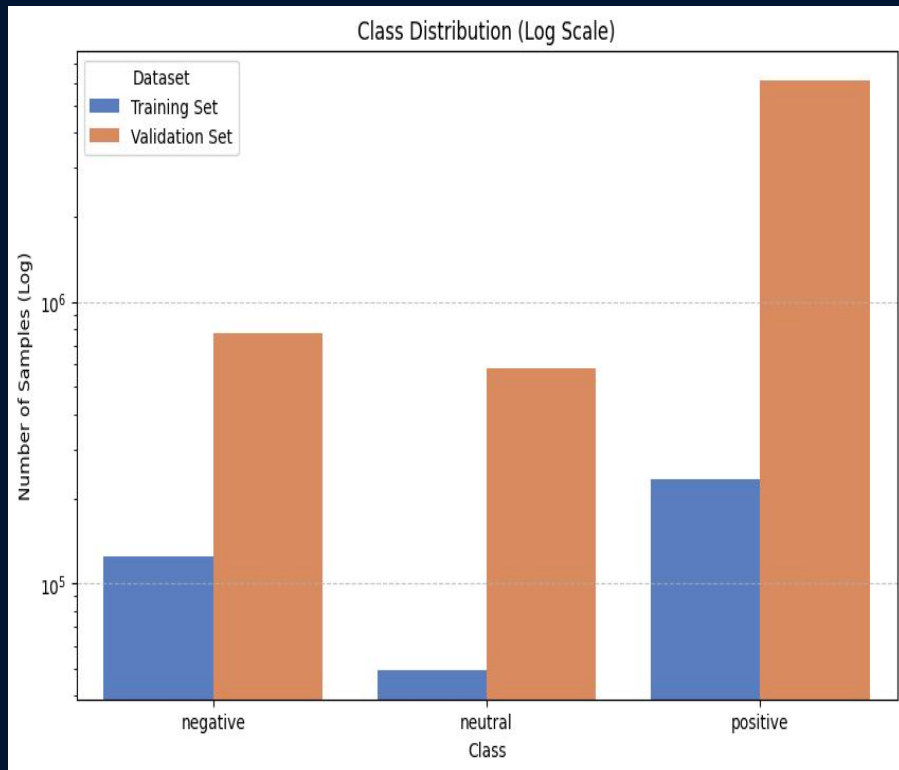
Key Challenge 2

Why standard SVM fails?!

- ❑ Optimizes global accuracy
- ❑ Biased toward majority class
- ❑ Ignores neutral sentiment features

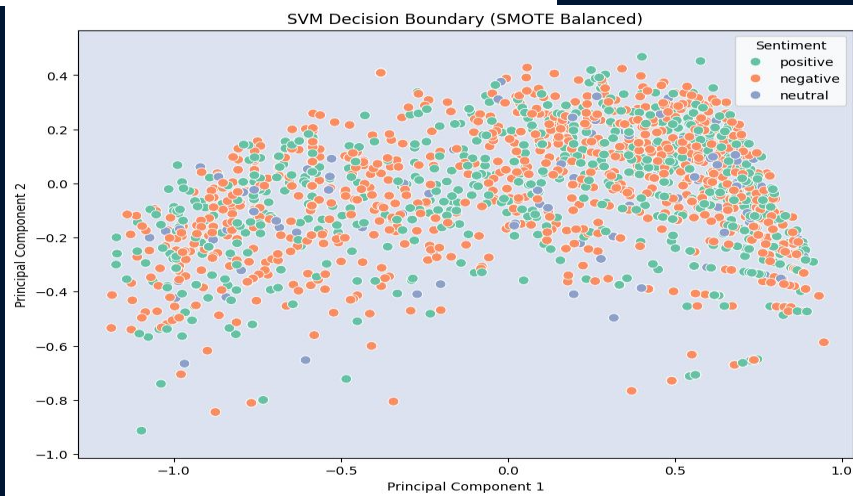
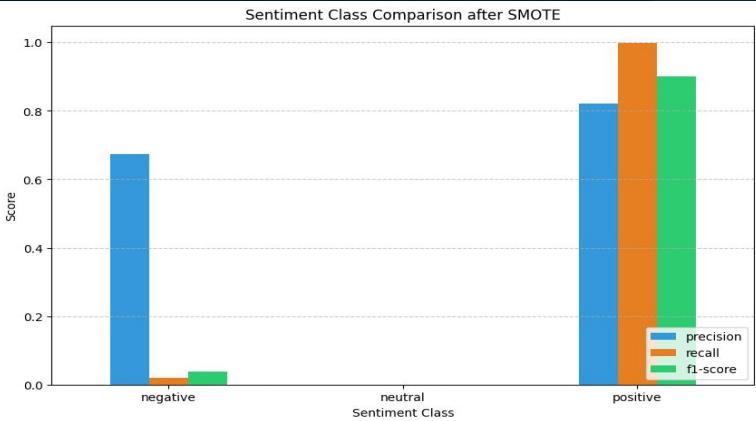


Cost-Sensitive SVM (CSV_COST)



- **Added class weights**
(`class_weight = balanced`)
- Penalized minority class errors
- Improvement:
 - Negative class performance increased
- **Limitation:**
 - Neutral class still not recovered
 -
 -

SMOTE (Data Balancing)



Synthetic Minority Over-sampling Technique

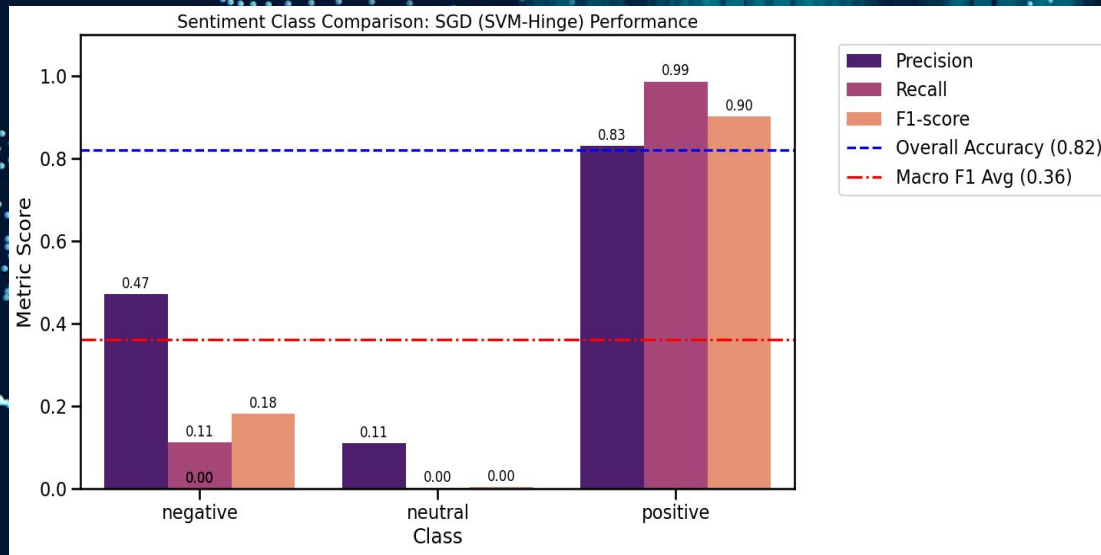
Generates artificial Neutral samples

Impact:

First recovery of Neutral Recall (0.04)

Insight:

Data imbalance was a core issue

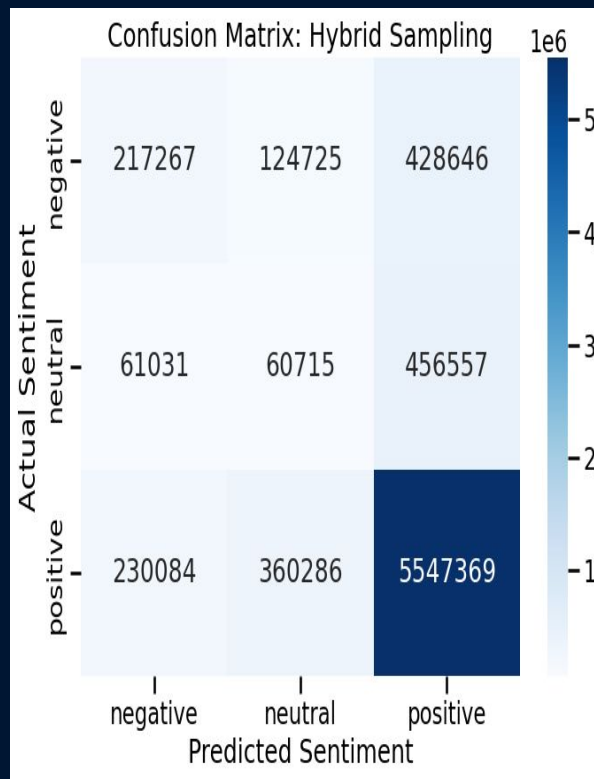
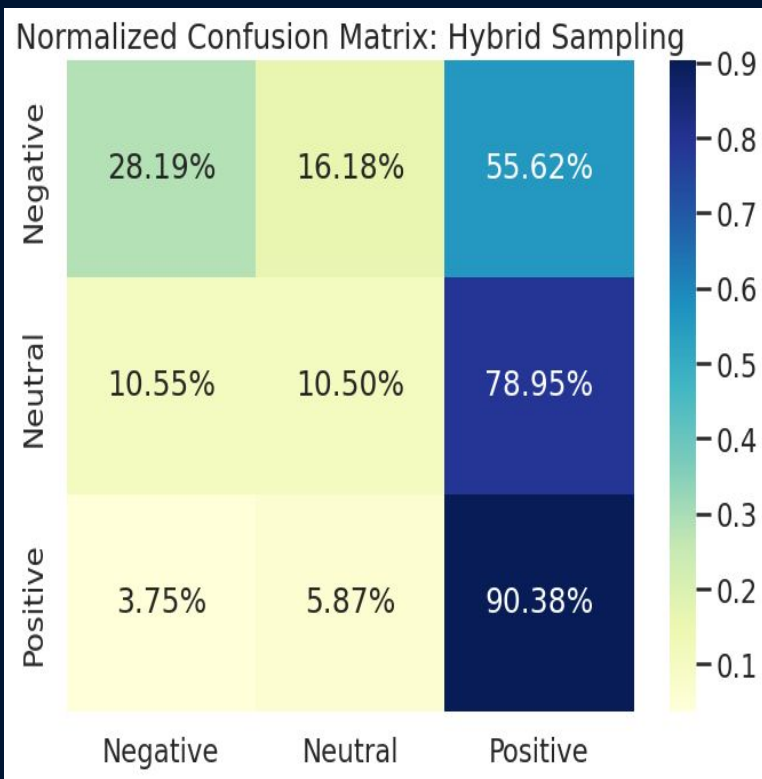


SGD for Scalability

- Stochastic Gradient Descent with hinge loss
- Handles millions of samples efficiently
- Training time reduced:
 - Hours → Minutes
- Maintains SVM-like decision boundaries

Final Hybrid Model

The "Neutral Recall" Breakthrough Indicates Successful Feature Engineering



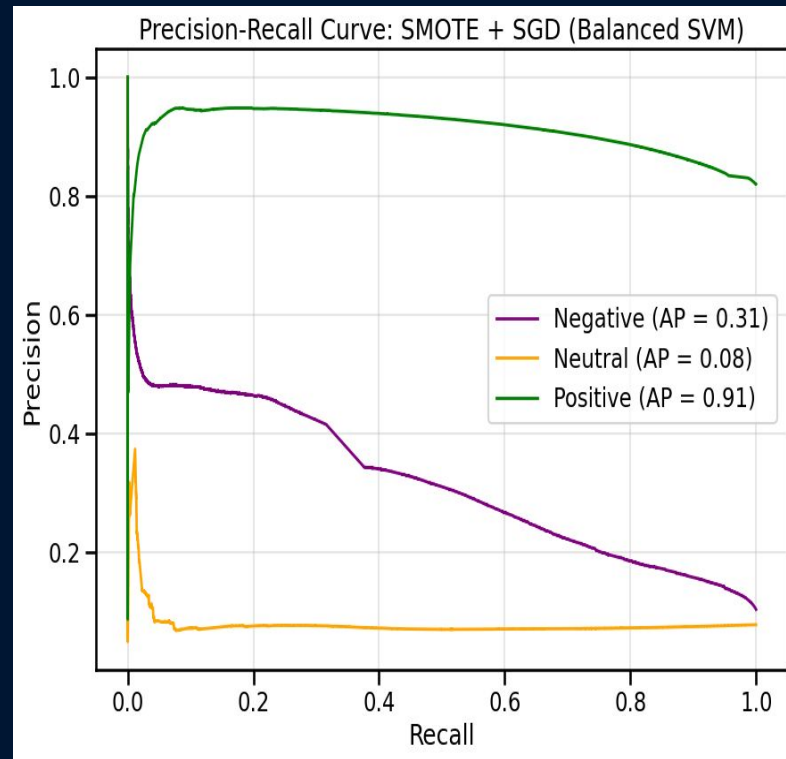
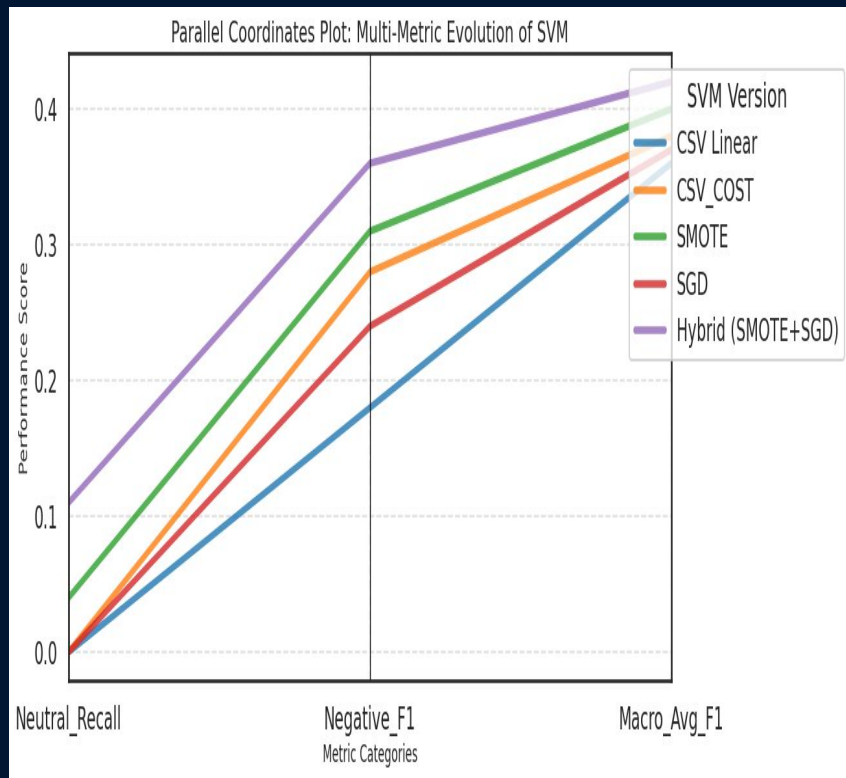
➤ **SMOTE + SGD + Hybrid Features**

TF-IDF + Sentiment Lexicons
Best overall balance

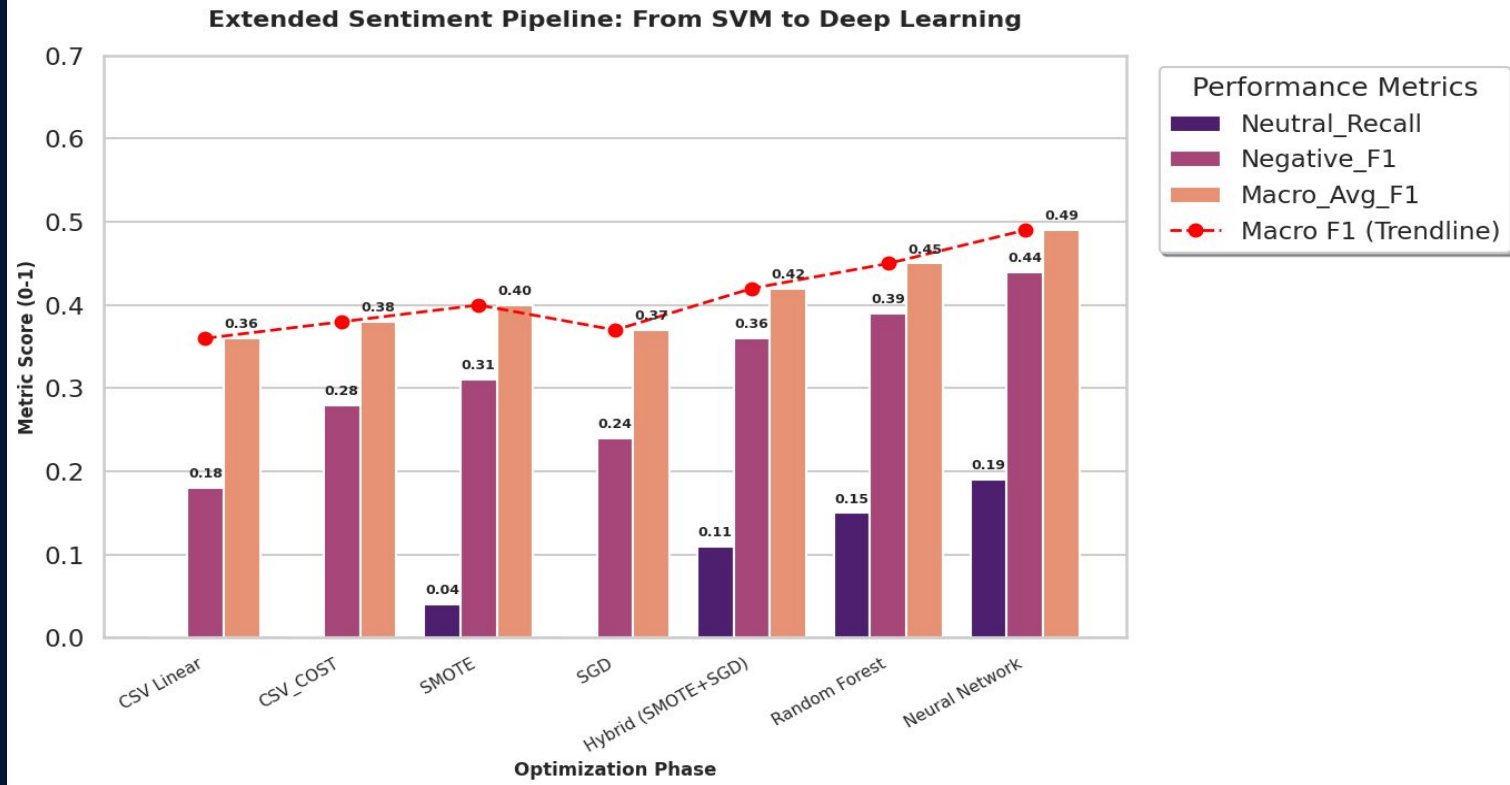
➤ **Results:**

- Neutral Recall: **0.11**
- Macro-F1: **0.42**

the performance of different sentiment classification models using Neutral Recall and Macro-F1 score,



Extended Sentiment Pipeline :From SVM to Deep Learning



Evaluation Strategy

Focus on Fairness, Not Accuracy

- ❑ Macro-Averaged F1 Score
- ❑ Normalized Confusion Matrices
- ❑ Prevents majority class dominance in evaluation

Key Findings

- Standard SVM is **blind to Neutral sentiment**
- SMOTE enables minority class visibility
- SGD ensures **scalability**
- Hybrid model balances:
 - Performance
 - Fairness
 - Computational efficiency

Conclusion

Class imbalance severely degrades sentiment models

Explicit balancing + scalable optimization is essential

Hybrid SMOTE + SGD:

- Improves fairness
- Maintains scalability

Strong alternative to deep learning for big data

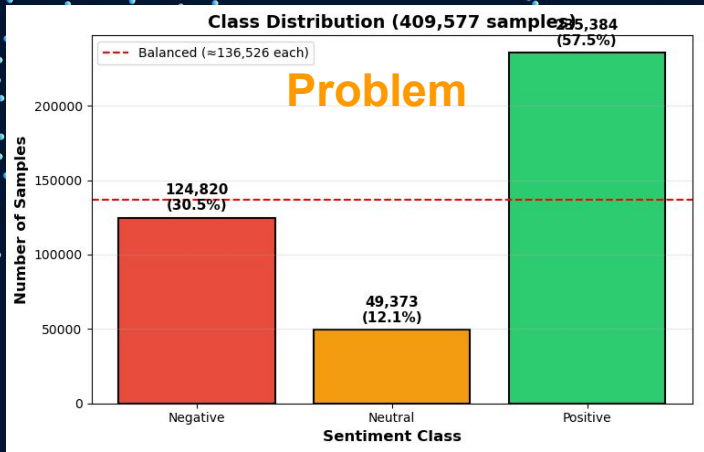
05

Neural Networks and Ensemble Methods

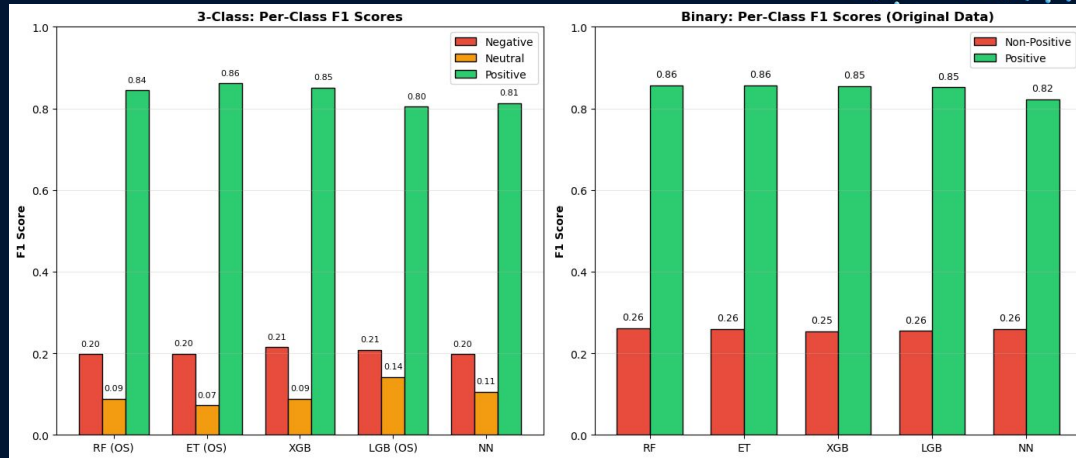
Noha Ismail

Methodology

Model	Type	Category
Random Forest	Supervised	Ensemble (Bagging)
Extra Trees	Supervised	Ensemble (Bagging)
XGBoost	Supervised	Ensemble (Boosting)
LightGBM	Supervised	Ensemble (Boosting)
Neural Network (MLP)	Supervised	Deep Learning
K-Means	Unsupervised	Clustering
DBSCAN	Unsupervised	Clustering



Initial Results





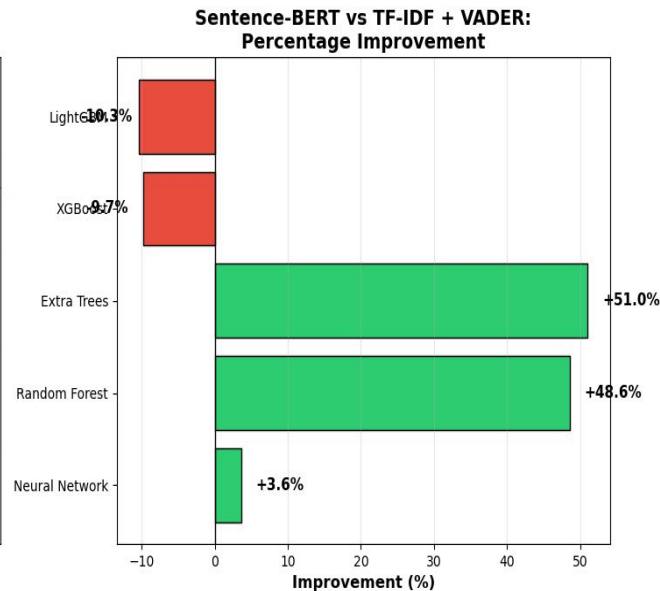
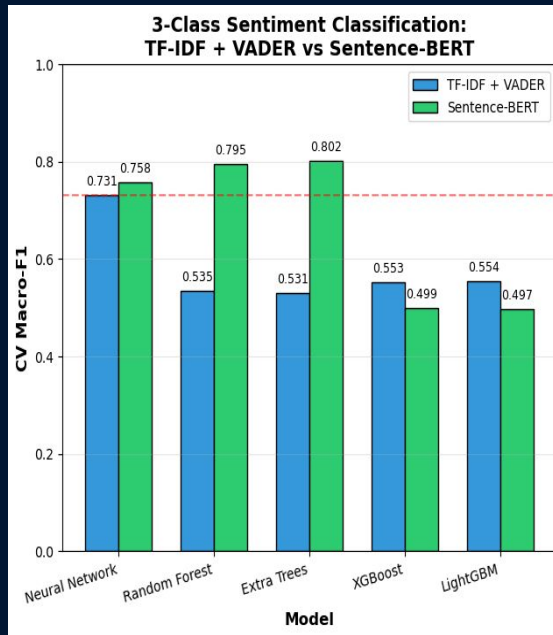
Let's divide our problem

- (1) How do SBERT embeddings compare to TF-IDF features for sentiment classification?
- (2) Why do certain algorithms fail on imbalanced sentiment data?
- (3) Do sentiment classes form natural clusters in the embedding space?

Transformer-based embeddings that capture contextual and semantic information



For computational resources these models were tested on subset of the dataset

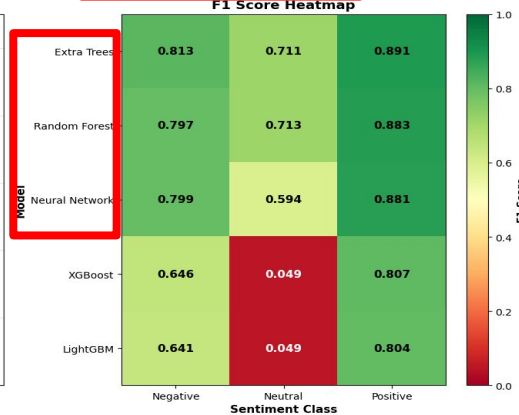
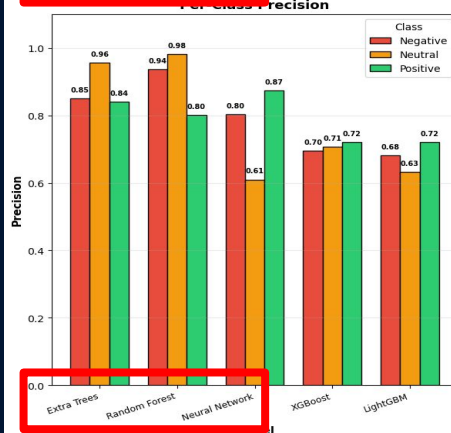
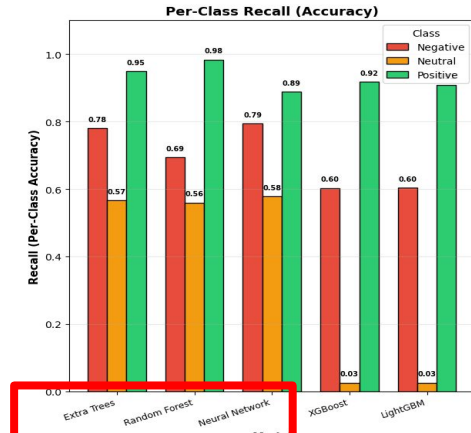
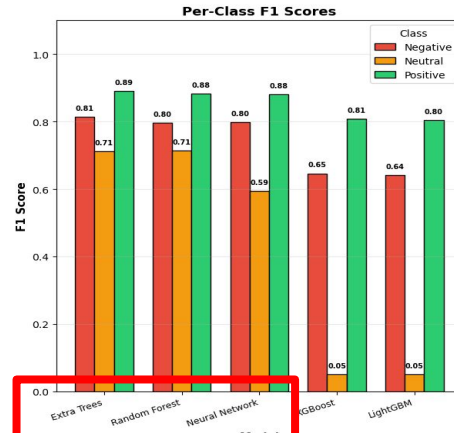
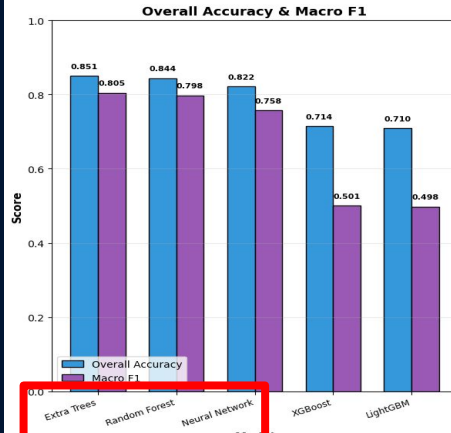


**For
computational
resources
these models
were tested on
subset of the
dataset**

Five classification algorithms were evaluated:

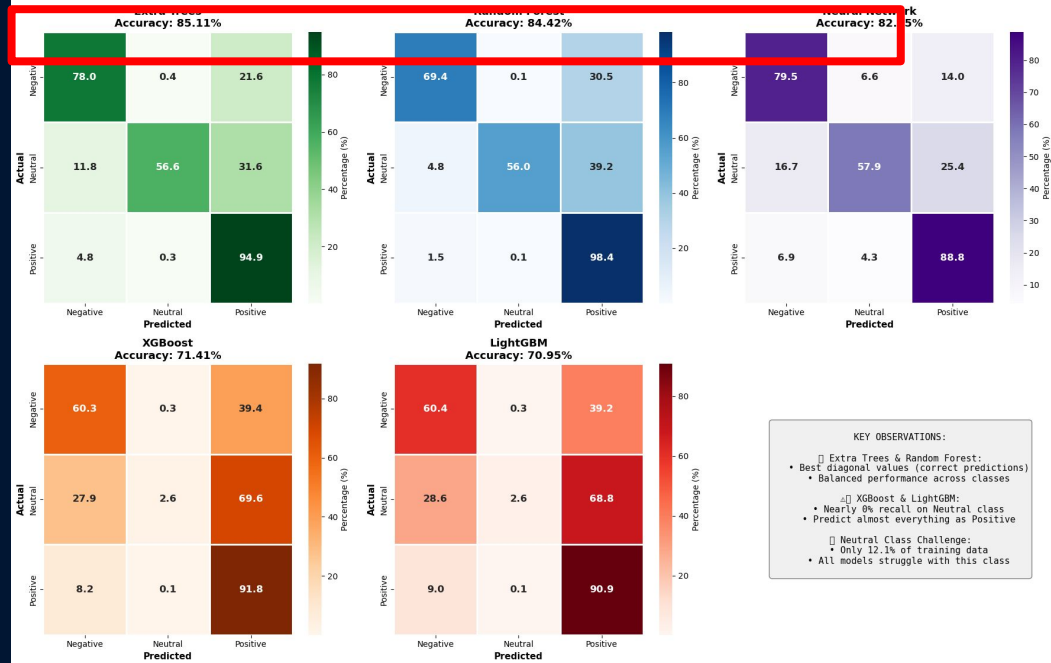
1. **Extra Trees Classifier:** `n_estimators=100, max_depth=20, class_weight='balanced'`
2. **Random Forest:** `n_estimators=200, max_depth=30, class_weight='balanced'`
3. **Neural Network (MLP):** `hidden_layers=(512, 256, 128), early_stopping=True`
4. **XGBoost:** `n_estimators=200, GPU-accelerated`
5. **LightGBM:** `n_estimators=200, num_leaves=31`

Comprehensive Evaluation: Sentence-BERT Embeddings (3-Class Sentiment Classification)

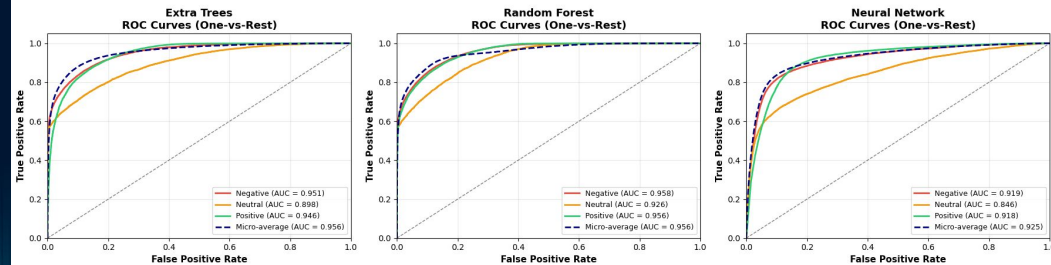


SENTENCE-BERT CLASSIFICATION SUMMARY	
□ BEST OVERALL ACCURACY:	Extra Trees: 85.11%
□ BEST MACRO F1:	Extra Trees: 0.8020
□ MODEL RANKING (by Accuracy):	
	1. Extra Trees - 85.11%
	2. Random Forest - 84.42%
	3. Neural Network - 82.25%
	4. XGBoost - 71.41%
	5. LightGBM - 70.95%
⚠️ NEUTRAL CLASS CHALLENGE:	
	• Best Neutral F1: Extra Trees (0.711)
	• Worst Neutral F1: XGBoost/LightGBM (0.049)
	• Class Imbalance: Neutral only 12.1% of data

Confusion Matrices: Sentence-BERT Embeddings (Values shown as % of actual class)

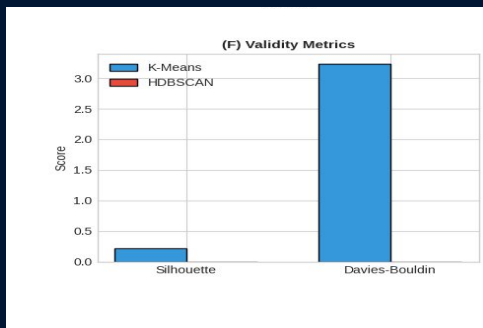
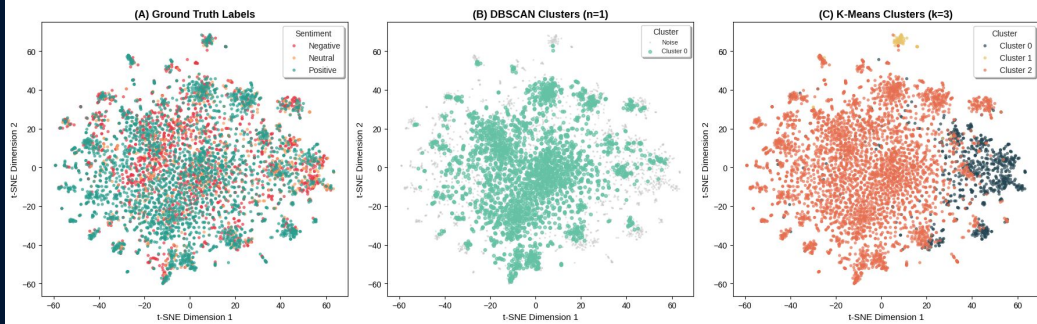


ROC Curves: Sentence-BERT Embeddings (3-Class Sentiment)



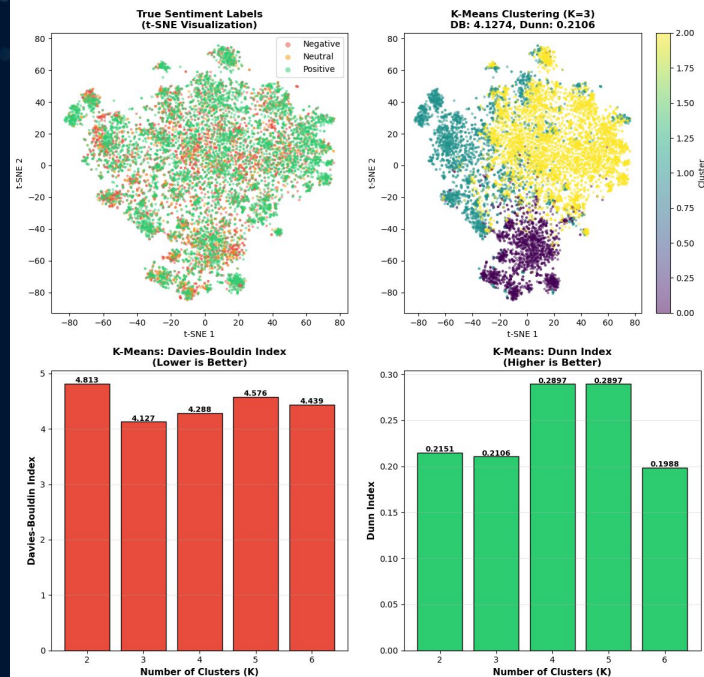
Using TF-IDF

Sentiment Analysis: Ground Truth vs Unsupervised Clustering



Using Sentence BERT

Clustering Analysis: Sentence-BERT Embedding (Dunn Index & Davies-Bouldin Index)



Conclusion

This study demonstrates that Superior BERT (80.14% accuracy), Extra Trees classifier achieved to the highest performance on key findings on (60.6%)

- Bagging methods (Extra Trees, Random Forest) showed dramatic improvement with SBERT embeddings (+48-51% over TF-IDF)
- Gradient boosting algorithms (XGBoost, LightGBM) failed on the minority Neutral class due to sequential learning bias and class imbalance sensitivity
- Dense semantic embeddings (384 dimensions) outperform sparse word-frequency vectors (3,004 dimensions) due to information density and contextual understanding
- Sentiment classes do not form natural clusters in embedding space, confirming that sentiment is a semantic rather than geometric property

Team Members:



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Ahmed Gomaa

THANKS!

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